

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
2			Method Diagram of charging circuit. Description of how to charge. Diagram of discharging circuit. Description of how to discharge. Collection of data described i.e. every 20 s measure V etc Valid meter correctly placed. Analysis A discharging equation given e.g. $I = I_0 e^{-\frac{t}{RC}}$ or $\ln V = \ln V_0 - \frac{t}{RC}$ Correct graph stated e.g. $I \propto t$, $V \propto t$, $Q \propto t$, $\ln V \propto t$. Valid method for time constant e.g. 37% left OR $\frac{-1}{\text{gradient}}$ Shape of graph described / shown correctly (i.e. exponential decay shape OR straight and negative gradient). Statement that time constant = RC (for comparison etc)	6			6	1	6